

Image CNN

AI FT – MARCH EN • Group 4

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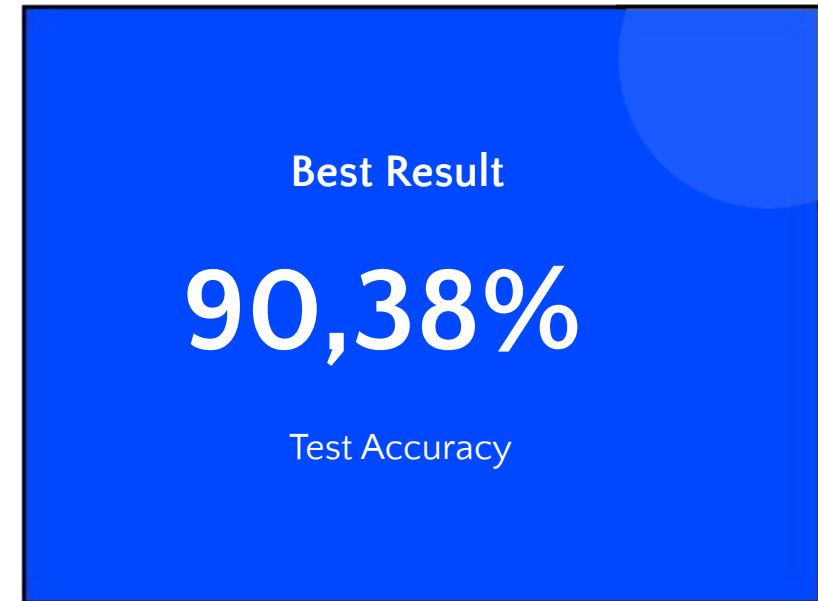


Agenda

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Executive Summary

- **Goal :** Accurate image classification with a compact CNN
- **Dataset:** CIFAR-10 (60,000 images, 10 classes)
- **Key Enablers:**
 - Architecture
 - Training
 - Regularization
- **Output:** Baseline-to-final model log and path to 90% via transfer learning and an interactive demo using **Gradio**



Dataset: CIFAR-10

- **Why CIFAR-10:**

- Standard benchmark for small-image classification
- Diverse natural images across 10 object categories
- Fast iteration for CNN prototyping

- **Data Facts:**

- 60,000 total images (32×32 RGB)
- 10 classes

- **What it looks like:**



Image Classification with Augmentation & Transfer Learning

Normalization

Scale pixel values to $[0,1]$ for stable optimization

Data Augmentation (Keras layers)

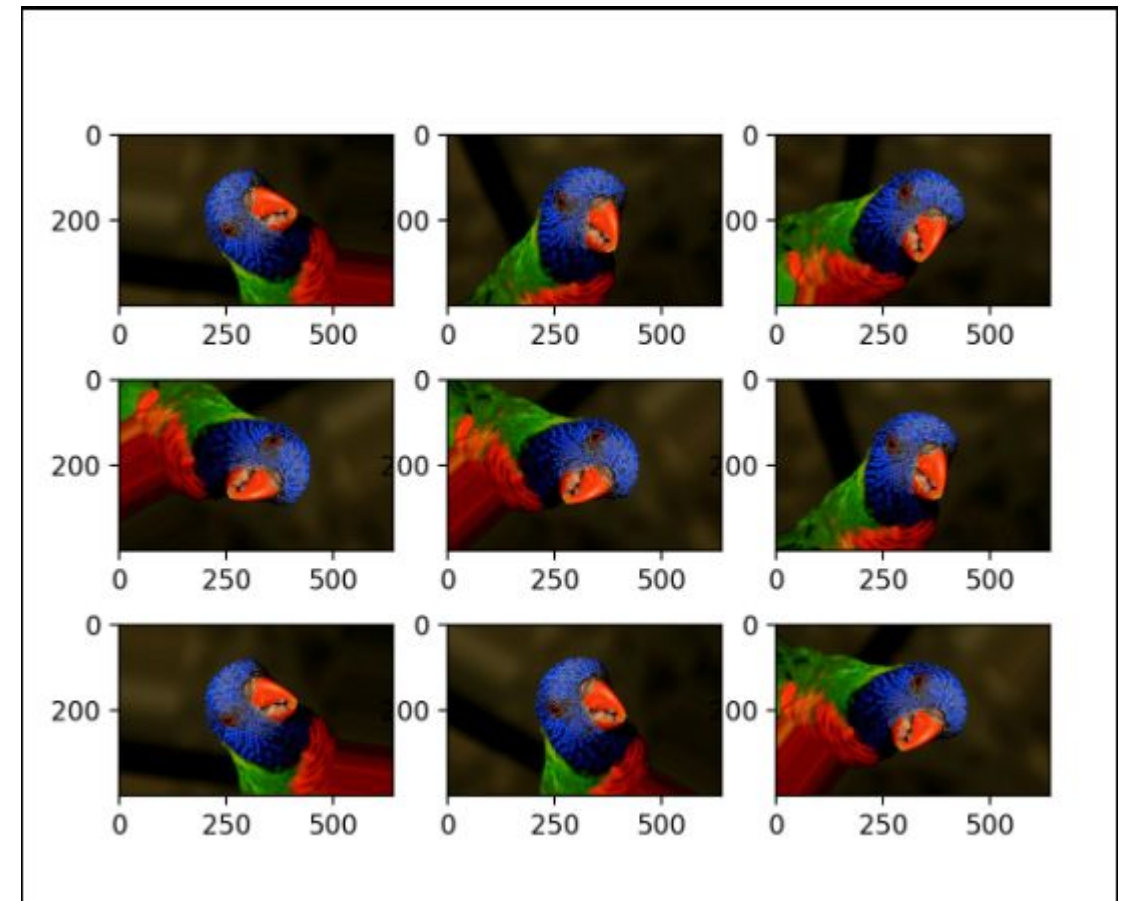
- `RandomFlip("horizontal")`
- `RandomRotation(0.1)`
- `RandomZoom(0.1)`

MobileNetV2 Transfer Learning

Architecture: GlobalAveragePooling (vs Flatten), 128 units in Dense, same Padding

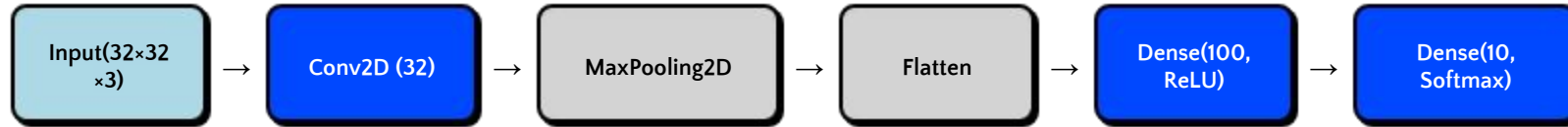
Training: Fine-tuned MobileNetV2, low Learning Rate, smaller Batch Size

Regularization: Dropout and Batch Normalization for convergence



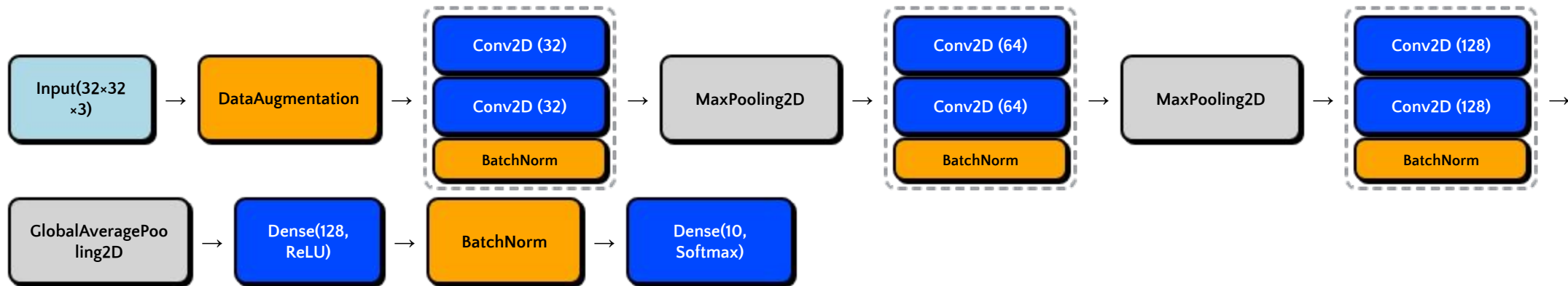
CNN Architecture without Transfer Learning

Base Model



Optimizer: SGD | LR = 0.001 | Batch size = 512 | Validation split = 0.2 | Epochs : 50

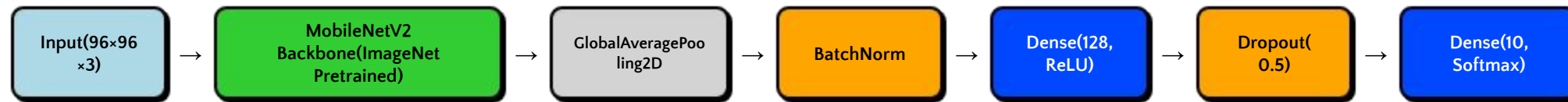
Best Model



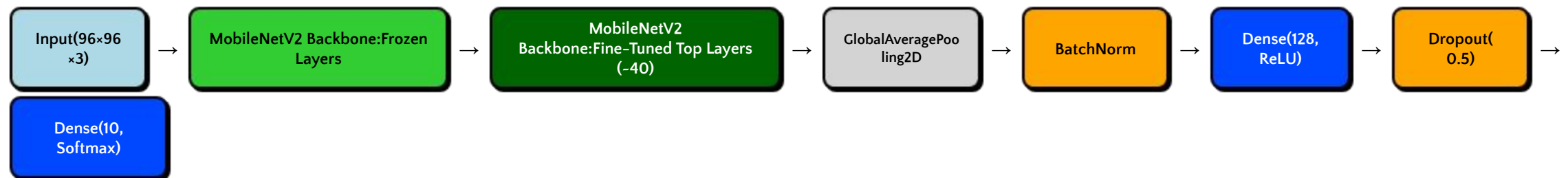
Optimizer: Adam | LR = 0.001 | Batch size = 64 | Validation split = 0.2 | Epochs : 50

CNN Architecture with Transfer Learning

Base Model



Best Model



Optimizer: Adam | LR = 1e-5 | Batch size = 16 | Validation split = 0.2 | Epochs : 10

Model Story & Optimization

Iteration Path

- Baseline CNN (low accuracy)
- Added more conv layers + adjusted filter sizes → better feature extraction + Transfer Learning

Learning Rate

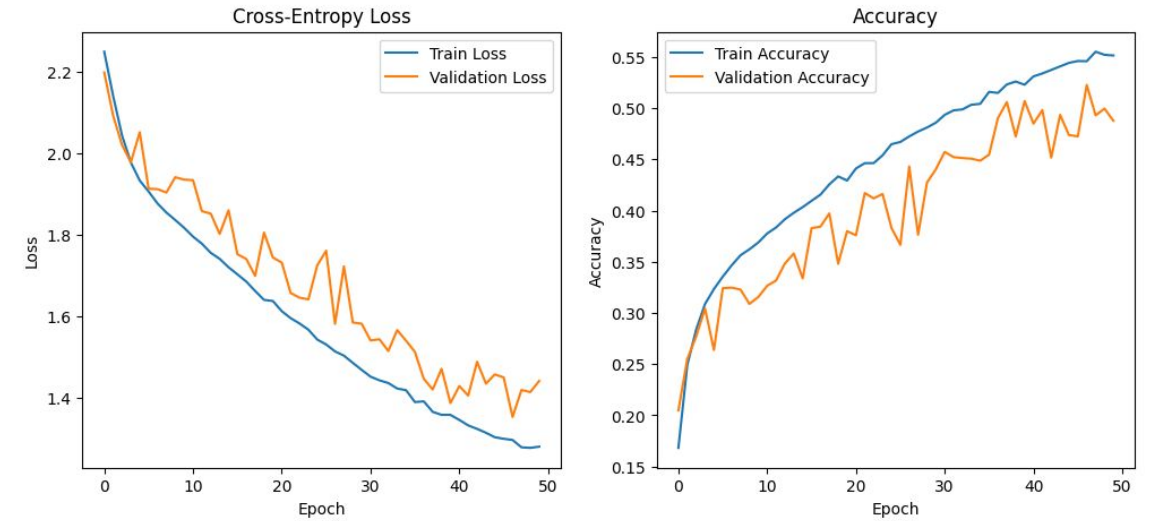
- Data augmentation and using Transfer Learning with lower learning rate had maximal impact

Breakthroughs

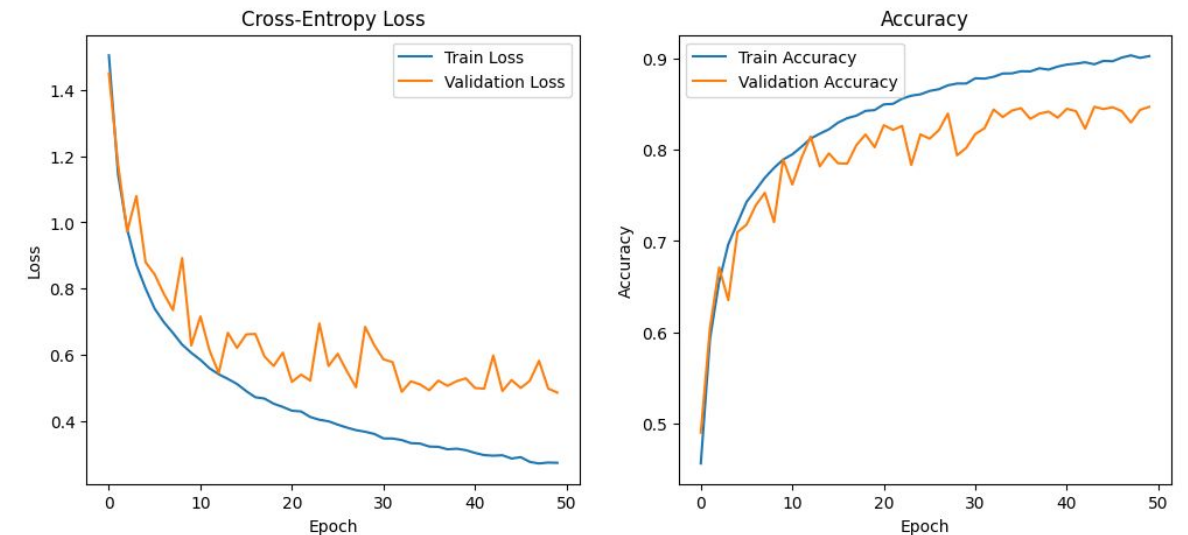
- 01** Data augmentation enabled high performance (Models 10 & 11) for models without Transfer Learning
- 02** Deeper network (6 conv layers in top models) lifted accuracy into 82%+
- 03** Batch Normalization + Dropout reduced overfitting

Learning Curve with No Transfer Learning (Toward 85%)

- **Goal:** Significant improvement in performance, Improve generalization, Reduce underfitting (low accuracy, high loss) and push toward 85% accuracy
- **Backbone:**
 - Base model: SGD
 - Tuned model: Adam



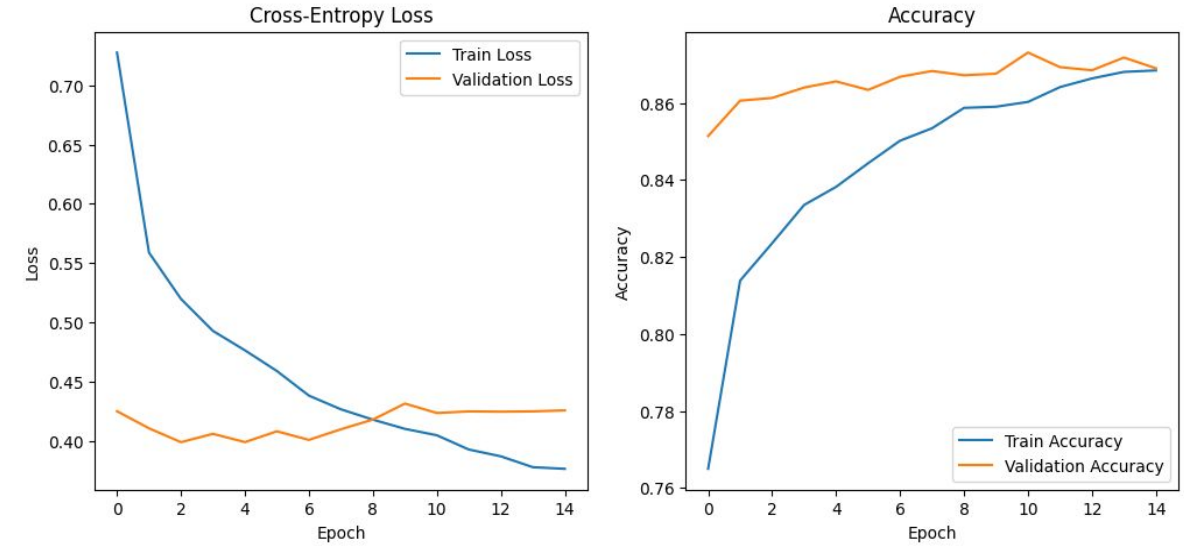
Base CNN Model



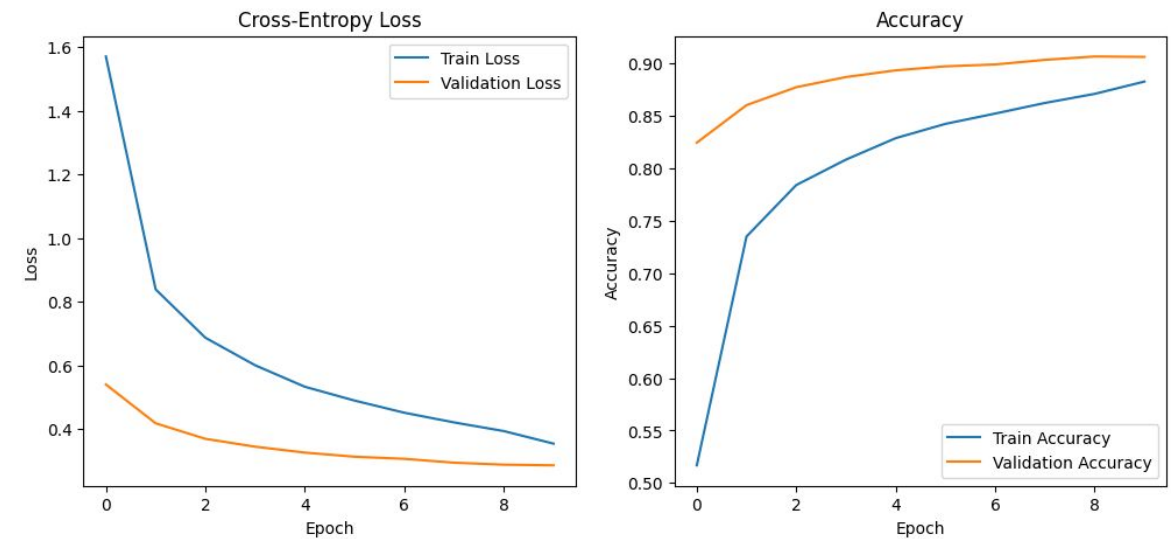
Tuned Model with Data Augmentation

Learning Curve with Transfer Learning (Toward 90%)

- **Goal:** Improve generalization, reduce overfit and push toward +90% accuracy
- **Backbone:** MobileNetV2



Base Transfer Learning Model



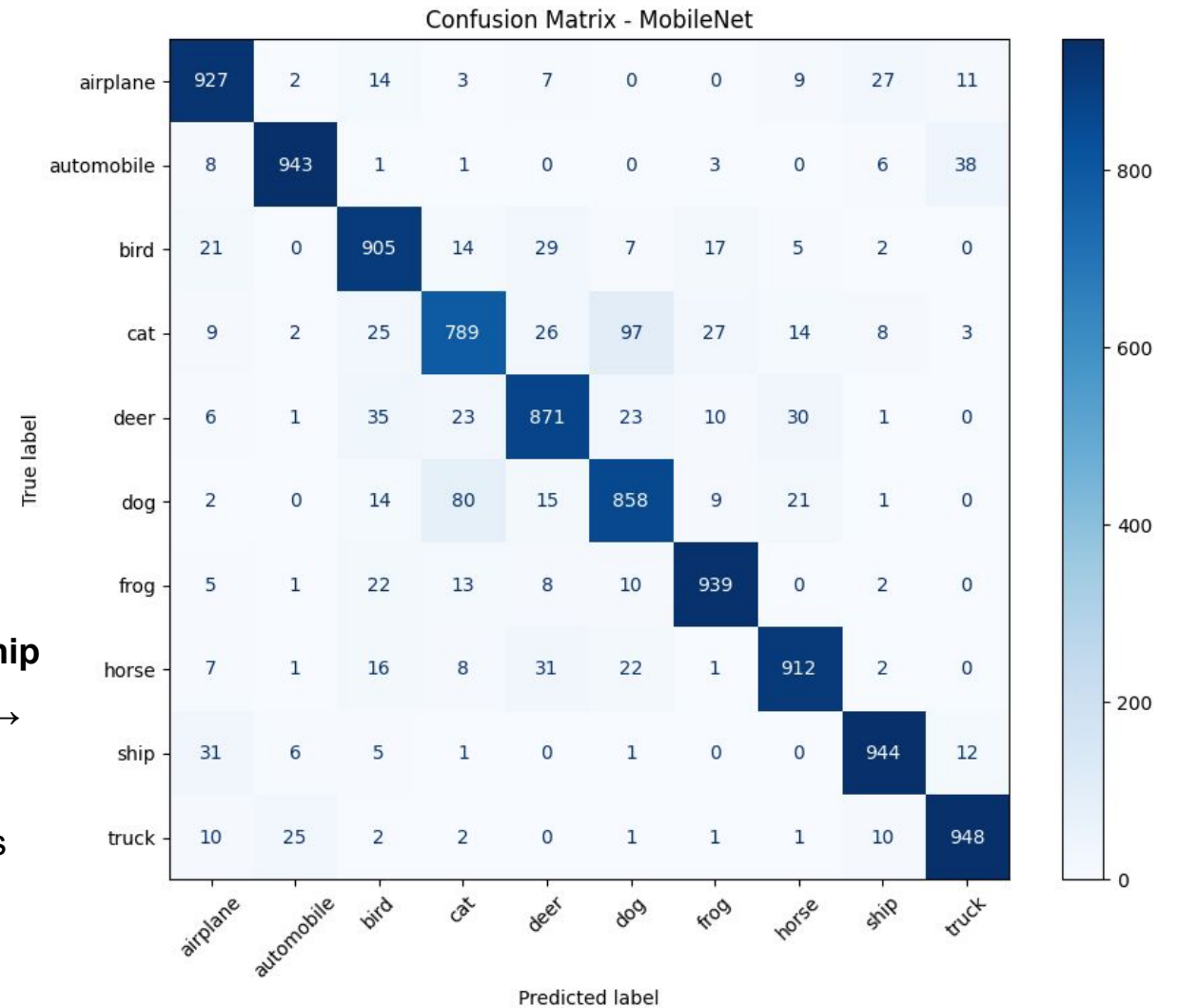
Tuned Transfer Learning Model

Confusion Metrics

Best Test Accuracy

90,38%

- High overall accuracy with most predictions on the diagonal (~900+ correct per class)
- Errors mainly occur between **visually similar classes**
- Largest confusions: **cat ↔ dog**, **automobile ↔ truck**, **airplane ↔ ship**
- Some confusion due to **background and low resolution** (e.g., bird ↔ deer)
- Model is **well-trained with strong generalization**, not random errors



Misclassified Images by True Class

- Misclassifications are mostly between visually similar classes and low-resolution ambiguous images.
- Errors are reasonable and not random, showing that the model has learned strong and meaningful features.
- Remaining mistakes are mainly caused by CIFAR-10's small image size, background effects, and overlap between similar animal and vehicle classes.

True: airplane
Pred: truck



True: automobile
Pred: truck



True: bird
Pred: dog



True: cat
Pred: deer



True: deer
Pred: bird



True: dog
Pred: deer



True: airplane
Pred: ship



True: automobile
Pred: truck



True: bird
Pred: airplane



True: cat
Pred: dog



True: deer
Pred: dog



True: dog
Pred: deer



True: airplane
Pred: ship



True: automobile
Pred: airplane



True: bird
Pred: horse



True: cat
Pred: deer



True: deer
Pred: horse

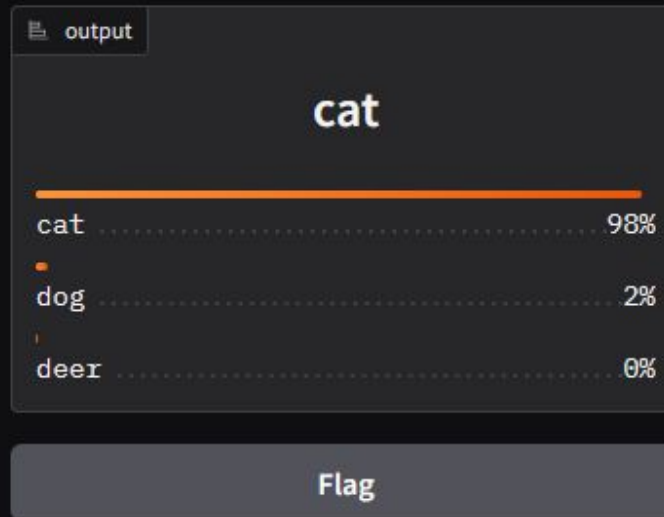
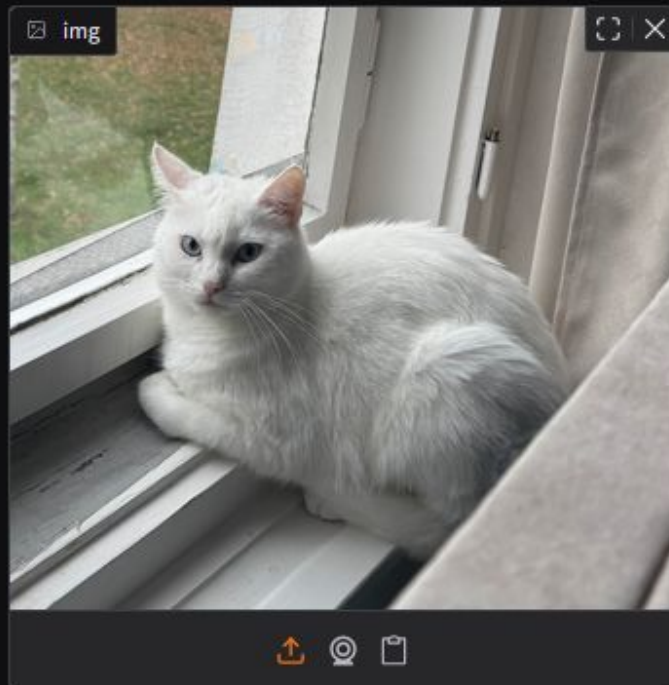


True: dog
Pred: horse



MobileNetV2 CIFAR-10 Classifier (Group 4)

Upload an image



Deployment & Key Learnings

- Data augmentation helps without transfer learning but can reduce accuracy with transfer learning.
- Depth + regularization (Dropout/BN) > LR tweaks for this project
- Clear iteration tracking speeds up model improvement